HK IMO Training 2023-2024
Problem Set 9 Suggested Solutions

Problem 33 (May Olympiad 2020 Level 2 Problem 2)**.**

(a) Does there exist positive integers a, b, c, d such that $a + b + c = 2020$ and $2^a + 2^b + 2^c = d^2$?

(b) Does there exist positive integers a, b, c, d such that $a + b + c = 2020$ and $3^a + 3^b + 3^c = d^2$?

Solution.

(a) Yes. We can take $(a, b, c) = (672, 674, 674)$ since $672 + 674 + 674 = 2020$ and

$$2^{672} + 2^{674} + 2^{674} = 2^{672}(1 + 4 + 4) = 2^{672} \cdot 9 = (2^{336} \cdot 3)^2.$$

(b) No. If $a + b + c = 2020$, then exactly 0 or 2 of a, b, c are odd. If none of a, b, c is odd, then $3^a + 3^b + 3^c \equiv 1 + 1 + 1 = 3 \pmod 4$. If exactly 2 of a, b, c are odd, then $3^a + 3^b + 3^c \equiv 3 + 3 + 1 = 7 \equiv 3 \pmod 4$. In both cases, $3^a + 3^b + 3^c$ cannot be a perfect square as $d^2 \equiv 0, 1 \pmod 4$.

$\square$

Problem 34 (Pan-American Girls' MO 2021 Problem 1)**.** There is a circular chessboard with $n \geqslant 2$ cells arranged in a circle. We put down the numbers 1 to n in the cells in any order such that each number is used exactly once. Initially, we put a chess piece in the cell numbered 1. Every time if the chess piece lies in the cell numbered k, we move the chess piece k steps in the clockwise direction and repeat the same process. Find all possible values of n for which we can write down the numbers in some way such that every cell can be visited at least once after finitely many moves. (Note that the chess piece always visits the cell numbered 1 initially.)

Solution. We use (i, j) to denote the ith cell (taken modulo n) counted from the cell with number 1 in the clockwise direction, such that the number inside this cell is j. For example, $(0, 1)$ is the 0th cell with number 1.

Note that once the chess piece visits $(*, n)$, it will stay in that cell forever. Therefore, this cell must be the last cell visited by the chess piece. Also, the chess piece cannot visit any other cell more than once, since otherwise it will move in a certain cycle indefinitely, but this contradicts our observation that it will stay in $(*, n)$ at the end.

Therefore, the chess piece must visit each of the cells $(*, k)$ with $k = 1, 2, \ldots, n-1$ exactly once in the first $n - 1$ steps. In these $n - 1$ steps, it has travelled $1 + 2 + \cdots + (n-1) = \dfrac{n(n-1)}{2}$ cells in total, and so the last cell must be $\left(\dfrac{n(n-1)}{2}, n\right)$. If n is odd, then this is the same as $(0, n)$ as $n \mid \dfrac{n(n-1)}{2}$. However, this cannot happen as the 0th cell contains the number 1. Thus, n must be even.

On the other hand, all even $n = 2m$ are possible. For example, suppose the cells are

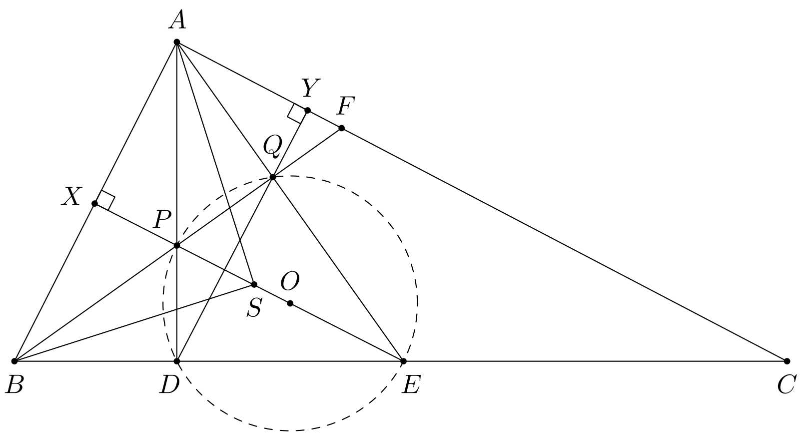
$$(0, 1), (1, 2m-2), (2, 2m-4), \ldots, (m-1, 2), (m, 2m), (m+1, 2m-1), (m+2, 2m-3), \ldots, (2m-1, 3).$$

Then it is easy to check that the chess piece will visit the cells

$$(0, 1), (1, 2m-2), (2m-1, 3), (2, 2m-4), (2m-2, 5), \ldots, (m-1, 2), (m+1, 2m-1), (m, 2m), (m, 2m), \ldots$$

in order. The condition is satisfied. $\square$

Problem 35 (Pan-American Girls' MO 2022 Problem 3)**.** In acute $\triangle ABC$, $AB < AC$. Let D and E be points on the side BC such that D lies between B and E, and $\angle BAD = \angle CAE$. Let F be a point on the side AC. The line BF intersects AD and AE at P and Q respectively. It is known that $DQ \perp AC$ and $EP \perp AB$. Let S be the intersection point of the angle bisectors of $\angle BAC$ and $\angle CBF$. Prove that A, B, D, S, Q are concyclic.



Solution. Firstly, since $\angle PDQ = \angle ADY = 90^\circ - \angle CAD = 90^\circ - \angle BAE = \angle AEX = \angle QEP$, the points D, E, Q, P are concyclic. Let O be the centre of $(DEQP)$. By Brokard's theorem, we have $EO \perp AB$. As $EP \perp AB$, O lies on EP. Therefore, EP is a diameter of $(DEQP)$. Thus, $\angle PDE = \angle PQE = 90^\circ$. This implies $\angle ADB = 90^\circ = \angle AQB$, and hence A, B, D, Q are concyclic.

Finally, note that AS bisects $\angle DAQ$ and BS bisects $\angle DBQ$. As A, B, D, Q are concyclic, these two angle bisectors must intersect at the midpoint of arc DQ of $(ABDQ)$. In particular, this shows S lies on $(ABDQ)$ as desired. $\square$

Problem 36 (KöMaL Points Contest A. 852)**.**

(a) Prove that the inequality $(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n) > \frac{2}{9}n^3$ holds for all positive integers n and all pairwise distinct pairs of positive integers (a_i, b_i) (i.e. $(a_i, b_i) \neq (a_j, b_j)$ for $i \neq j$).

(b) Prove that the constant $\frac{2}{9}$ in part (a) cannot be replaced by any larger constant.

Solution.

(a) Let c_j be the number of pairs (a_i, b_i) with $a_i = j$. Clearly, we may replace these pairs by $(j, 1), (j, 2), \ldots, (j, c_j)$ to decrease the size of the left-hand side of the inequality. After this replacement, we have

$$b_1 + b_2 + \cdots + b_n = \sum_{j=1}^{k} \sum_{i=1}^{c_j} i = \sum_{j=1}^{k} \frac{c_j(c_j+1)}{2} > \frac{1}{2} \sum_{j=1}^{k} c_j^2,$$

for some k. Therefore, it suffices to prove

$$(c_1 + 2c_2 + \cdots + kc_k)(c_1^2 + c_2^2 + \cdots + c_k^2) \geqslant \frac{4}{9}n^3.$$

Let $x_j = \frac{c_j}{n}$ for each j. Note that $x_1 + x_2 + \cdots + x_k = \frac{c_1 + c_2 + \cdots + c_k}{n} = 1$. It remains to prove

$$(x_1 + 2x_2 + \cdots + kx_k)(x_1^2 + x_2^2 + \cdots + x_k^2) \geqslant \frac{4}{9} \tag{1}$$

under the constraints $x_1 + x_2 + \cdots + x_k = 1$ and $x_j \geqslant 0$.

For each fixed k, the left-hand side of (1) must attain a minimum value by the extreme value theorem. This minimal case must satisfy $x_1 \geqslant x_2 \geqslant \cdots \geqslant x_k$ by the rearrangement inequality. To prove (1), we may also assume $x_k > 0$ since otherwise we can remove the term x_k (i.e. reducing the size of k). Thus, we can use the method of Lagrange multiplier. The minimal case must satisfy

$$\begin{aligned}
(x_1^2 + x_2^2 + \cdots + x_k^2) + 2x_1(x_1 + 2x_2 + \cdots + kx_k) &= \lambda, \\
2(x_1^2 + x_2^2 + \cdots + x_k^2) + 2x_2(x_1 + 2x_2 + \cdots + kx_k) &= \lambda, \\
&\cdots, \\
k(x_1^2 + x_2^2 + \cdots + x_k^2) + 2x_k(x_1 + 2x_2 + \cdots + kx_k) &= \lambda
\end{aligned}$$

for some λ. By considering the difference of every pair of consecutive equations, we see that $x_1, x_2, \ldots, x_k$ form an arithmetic progression.

Let $x_j = t + (k-j)d$ for $j = 1, 2, \ldots, k$. The condition $x_1 + x_2 + \cdots + x_k = 1$ becomes $kt + \frac{k(k-1)}{2}d = 1$, and hence

$$d = \frac{2(1-kt)}{k(k-1)}.$$

To prove (1), it suffices to prove

$$\left(\sum_{j=1}^{k} j[t+(k-j)d]\right)\left(\sum_{j=1}^{k}[t+(k-j)d]^2\right) \geqslant \frac{4}{9},$$

or

$$\left(\frac{k(k+1)(t+kd)}{2} - \frac{k(k+1)(2k+1)d}{6}\right)\left(k(t+kd)^2 - k(k+1)d(t+kd) + \frac{k(k+1)(2k+1)d^2}{6}\right) \geqslant \frac{4}{9}.$$

We put $d = \frac{2(1-kt)}{k(k-1)}$. The first bracket is

$$\frac{k(k+1)}{2}t + \frac{k(k+1)(k-1)}{6}d = \frac{k(k+1)}{2}t + \frac{(k+1)(1-kt)}{3} = \frac{k+1}{3} + \frac{k(k+1)}{6}t$$

while the second bracket is

$$\begin{aligned} kt^2 + k(k-1)td + \frac{k(k-1)(2k-1)}{6}d^2 &= kt^2 + 2(1-kt)t + \frac{2(2k-1)(1-kt)^2}{3k(k-1)} \\ &= \frac{2(2k-1)}{3k(k-1)} - \frac{2(k+1)}{3(k-1)}t + \frac{k(k+1)}{3(k-1)}t^2. \end{aligned}$$

Expanding the bracket, it remains to prove

$$\frac{4(k+1)(2k-1)}{k(k-1)} - \frac{6(k+1)}{k-1}t + \frac{k^2(k+1)^2}{k-1}t^3 \geqslant 8,$$

or

$$k^3(k+1)^2t^3 - 6k(k+1)t + 4(3k-1) \geqslant 0.$$

Let $f(t) = k^3(k+1)^2t^3 - 6k(k+1)t + 4(3k-1)$. Then $f'(t) = 3k^3(k+1)^2t^2 - 6k(k+1)$. Solving $f'(t) = 0$, we obtain $t = t_0 = \sqrt{\frac{2}{k^2(k+1)}}$ as $t = x_k \geqslant 0$. Clearly, $f'(t) < 0$ when $t < t_0$ and $f'(t) > 0$ when $t > t_0$. Therefore, f attains its minimum at $t = t_0$. In that case, we have

$$f(t_0) = -4\sqrt{2(k+1)} + 4(3k-1).$$

Note that $f(t_0) \geqslant 0$ iff $3k-1 \geqslant \sqrt{2(k+1)}$ iff $9k^2 - 8k - 1 \geqslant 0$, or $(k-1)(9k+1) \geqslant 0$. This proves the inequality.

(b) Let m be a sufficiently large positive integer. Consider all pairs

$$(a_j, b_j) = (1,1), (1,2), (2,1), (1,3), (2,2), (3,1), \ldots, (1, m-1), (2, m-2), \ldots, (m-1, 1)$$

of positive integers with sum at most m. There are $n = 1 + 2 + \cdots + (m-1) = \frac{m(m-1)}{2}$ pairs in total. Also, we have

$$a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_n = \sum_{j=1}^{m-1}(m-j)j = \frac{m(m-1)(m+1)}{6}.$$

Therefore, we have

$$\frac{(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n)}{n^3} = \frac{2(m+1)^2}{9m(m-1)}.$$

When $m \to \infty$, this tends to $\frac{2}{9}$. Thus, the constant $\frac{2}{9}$ cannot be replaced by any larger constant.

$\square$